

Recitation 3

Gaussian Random Vectors and Bayesian Models Review and Practice Problems

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1 Gaussian random vectors

A d -dimensional Gaussian random vector with mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and positive-definite covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is written

$$\tilde{\mathbf{x}} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

Its joint pdf is

$$f_{\boldsymbol{\mu}, \boldsymbol{\Sigma}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d \det(\boldsymbol{\Sigma})}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

The diagonal entry $\boldsymbol{\Sigma}_{jj}$ is the variance of coordinate j , and $\boldsymbol{\Sigma}_{jk}$ is the covariance between coordinates j and k .

Every linear transformation of a Gaussian random vector is Gaussian:

$$\mathbf{A}\tilde{\mathbf{x}} + \mathbf{b} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Why? One characterization of a Gaussian random vector is that every linear combination of its coordinates is a univariate Gaussian random variable. Set $\tilde{\mathbf{z}} := \mathbf{A}\tilde{\mathbf{x}} + \mathbf{b}$. For any fixed vector $\mathbf{c}$,

$$\mathbf{c}^\top \tilde{\mathbf{z}} = (\mathbf{A}^\top \mathbf{c})^\top \tilde{\mathbf{x}} + \mathbf{c}^\top \mathbf{b}.$$

The right-hand side is a linear combination of the coordinates of $\tilde{\mathbf{x}}$ plus a constant, so it is Gaussian. Since this holds for every $\mathbf{c}$, $\tilde{\mathbf{z}}$ is a Gaussian random vector. Its parameters follow from linearity of the mean and the covariance rules:

$$\begin{aligned} \mathbb{E}[\tilde{\mathbf{z}}] &= \mathbf{A}\mathbb{E}[\tilde{\mathbf{x}}] + \mathbf{b} = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \\ \text{Cov}(\tilde{\mathbf{z}}) &= \mathbf{A} \text{Cov}(\tilde{\mathbf{x}}) \mathbf{A}^\top = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top. \end{aligned}$$

Consequently, every subvector is Gaussian. If

$$\begin{bmatrix} \tilde{\mathbf{x}} \\ \tilde{\mathbf{y}} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{bmatrix} \right),$$

then conditioning also preserves Gaussianity:

$$\tilde{\mathbf{y}} \mid \tilde{\mathbf{x}} = \mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_y + \boldsymbol{\Sigma}_{yx} \boldsymbol{\Sigma}_{xx}^{-1} (\mathbf{x} - \boldsymbol{\mu}_x), \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx} \boldsymbol{\Sigma}_{xx}^{-1} \boldsymbol{\Sigma}_{xy}).$$

Why? Define

$$\mathbf{B} := \Sigma_{yx} \Sigma_{xx}^{-1}, \quad \tilde{\mathbf{r}} := \tilde{\mathbf{y}} - \boldsymbol{\mu}_y - \mathbf{B}(\tilde{\mathbf{x}} - \boldsymbol{\mu}_x).$$

The pair $(\tilde{\mathbf{x}}, \tilde{\mathbf{r}})$ is jointly Gaussian because it is a linear transformation of $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$. Moreover,

$$\text{Cov}(\tilde{\mathbf{r}}, \tilde{\mathbf{x}}) = \Sigma_{yx} - \mathbf{B} \Sigma_{xx} = \mathbf{0}.$$

For jointly Gaussian vectors, zero covariance implies independence, so $\tilde{\mathbf{r}}$ is independent of $\tilde{\mathbf{x}}$. Its mean is zero and its covariance is

$$\text{Cov}(\tilde{\mathbf{r}}) = \Sigma_{yy} - \Sigma_{yx} \Sigma_{xx}^{-1} \Sigma_{xy}.$$

Finally, rewrite $\tilde{\mathbf{y}}$ as

$$\tilde{\mathbf{y}} = \boldsymbol{\mu}_y + \mathbf{B}(\tilde{\mathbf{x}} - \boldsymbol{\mu}_x) + \tilde{\mathbf{r}}.$$

After conditioning on $\tilde{\mathbf{x}} = \mathbf{x}$, the first two terms are fixed and the independent Gaussian residual $\tilde{\mathbf{r}}$ is unchanged. This gives exactly the conditional mean and covariance above.

2 Maximum likelihood for a Gaussian mean vector

Suppose

$$\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \Sigma),$$

where Σ is fixed and positive definite. For the observed dataset $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, the terms of the log-likelihood that depend on $\boldsymbol{\mu}$ are

$$\ell(\boldsymbol{\mu}) = C - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}_i - \boldsymbol{\mu}).$$

The gradient and Hessian are

$$\nabla_{\boldsymbol{\mu}} \ell(\boldsymbol{\mu}) = \Sigma^{-1} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}), \quad \nabla_{\boldsymbol{\mu}}^2 \ell(\boldsymbol{\mu}) = -n \Sigma^{-1}.$$

Because Σ^{-1} is positive definite, the Hessian is negative definite, so the unique maximizer is

$$\boldsymbol{\mu}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i.$$

Thus the ML estimate is obtained by averaging each coordinate.

3 Bayesian parametric models

In a Bayesian model, the parameter is itself random. Let $\tilde{\theta}$ be a parameter and let $\tilde{\mathbf{x}}$ represent the data. The model specifies

prior:	$f_{\tilde{\theta}}(\theta),$
likelihood:	$f_{\tilde{\mathbf{x}} \tilde{\theta}}(\mathbf{x} \mid \theta),$
posterior:	$f_{\tilde{\theta} \tilde{\mathbf{x}}}(\theta \mid \mathbf{x}) = \frac{f_{\tilde{\mathbf{x}} \tilde{\theta}}(\mathbf{x} \mid \theta) f_{\tilde{\theta}}(\theta)}{f_{\tilde{\mathbf{x}}}(\mathbf{x})}.$

The denominator is a normalizing constant, so as a function of θ ,

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

Two conjugate updates will be useful today. A prior is called **conjugate** when the posterior belongs to the same distribution family as the prior.

Gaussian prior and Gaussian likelihood. Suppose

$$\tilde{\mu} \sim \mathcal{N}(m_0, \tau^2), \quad \tilde{y} \mid \tilde{\mu} = \mu \sim \mathcal{N}(\mu, s^2),$$

where τ^2 describes prior uncertainty and s^2 describes sensor noise. Bayes' rule gives, as a function of μ ,

$$\begin{aligned} f_{\tilde{\mu}|\tilde{y}}(\mu \mid y) &\propto f_{\tilde{y}|\tilde{\mu}}(y \mid \mu) f_{\tilde{\mu}}(\mu) \\ &\propto \exp\left[-\frac{(y - \mu)^2}{2s^2}\right] \exp\left[-\frac{(\mu - m_0)^2}{2\tau^2}\right]. \end{aligned}$$

Expanding the two squares and keeping the terms that involve μ gives

$$\log f_{\tilde{\mu}|\tilde{y}}(\mu \mid y) = -\frac{1}{2} \left[\left(\frac{1}{\tau^2} + \frac{1}{s^2} \right) \mu^2 - 2 \left(\frac{m_0}{\tau^2} + \frac{y}{s^2} \right) \mu \right] + C.$$

Define

$$v_1 := \left(\frac{1}{\tau^2} + \frac{1}{s^2} \right)^{-1}, \quad m_1 := v_1 \left(\frac{m_0}{\tau^2} + \frac{y}{s^2} \right).$$

Completing the square turns the preceding expression into

$$-\frac{(\mu - m_1)^2}{2v_1} + C',$$

which is the log-pdf of a Gaussian distribution. Therefore

$$\boxed{\tilde{\mu} \mid \tilde{y} = y \sim \mathcal{N}(m_1, v_1).}$$

The quantities $1/\tau^2$ and $1/s^2$ are called **precisions**. The posterior precision is their sum, and

$$m_1 = \frac{s^2}{s^2 + \tau^2} m_0 + \frac{\tau^2}{s^2 + \tau^2} y$$

is a weighted average of the prior mean and the observation. A more precise source receives more weight. Also $v_1 < \tau^2$ and $v_1 < s^2$, so combining the prior with the observation reduces uncertainty.

Beta prior and Bernoulli likelihood. Suppose $\tilde{\theta} \sim \text{Beta}(a, b)$, so on $0 < \theta < 1$ its prior density has the form

$$f_{\tilde{\theta}}(\theta) \propto \theta^{a-1} (1 - \theta)^{b-1}.$$

Given $\tilde{\theta} = \theta$, let the Bernoulli observations be conditionally independent. If the observed data contain h heads and t tails, their likelihood is

$$p_{\tilde{x}|\tilde{\theta}}(\mathbf{x} \mid \theta) = \prod_i \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^h (1 - \theta)^t.$$

Multiplying likelihood and prior gives

$$\begin{aligned} f_{\tilde{\theta}|\tilde{x}}(\theta \mid \mathbf{x}) &\propto \theta^h (1 - \theta)^t \theta^{a-1} (1 - \theta)^{b-1} \\ &= \theta^{a+h-1} (1 - \theta)^{b+t-1}. \end{aligned}$$

This is the kernel of another beta density, so

$$\boxed{\tilde{\theta} \mid \text{data} \sim \text{Beta}(a + h, b + t).}$$

The update simply adds the observed counts to the two beta parameters. For example, the uniform prior is $\text{Beta}(1, 1)$; after two heads it becomes $\text{Beta}(3, 1)$.

Practice problems

Problems 2 and 3 are adapted from Exercises 5.15 and 6.14 of *Probability and Statistics for Data Science*.

Problem 1: MLE of a Gaussian mean vector

[6 minutes]

Let

$$\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \tilde{\mathbf{x}}_3 \stackrel{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

and suppose the observed vectors are

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

- (a) Write the log-likelihood as a function of $\boldsymbol{\mu}$, up to an additive constant.
- (b) Differentiate the general n -observation expression and derive $\boldsymbol{\mu}_{\text{ML}}$.
- (c) Compute $\boldsymbol{\mu}_{\text{ML}}$ for the observed dataset.
- (d) Would replacing $\boldsymbol{\Sigma}$ by the identity matrix change this ML estimate? Briefly explain.

Problem 2: Gaussian Bayesian model (Exercise 5.15)**[9 minutes]**

A noisy sensor is used to estimate the temperature on top of a mountain. Let $\tilde{\mu}$ denote the unknown temperature. The prior and sensor model are

$$\tilde{\mu} \sim \mathcal{N}(0, \sigma^2), \quad \tilde{y} \mid \tilde{\mu} = \mu \sim \mathcal{N}(\mu, 1).$$

- (a) Show that $[\tilde{\mu}, \tilde{y}]^\top$ is a Gaussian random vector with mean zero, and derive its covariance matrix.
- (b) Derive the posterior distribution of $\tilde{\mu}$ given $\tilde{y} = y$.
- (c) Let $y = -1$ and $\sigma^2 = 0.01$. Describe or sketch the prior and posterior. Does the prior express confidence that the temperature is close to 0? Does one reading change it much?
- (d) Repeat part (c) for $y = -1$ and $\sigma^2 = 10$.

Problem 3: Two coin flips (Exercise 6.14)**[7 minutes]**

Let the probability of heads be a random variable $\tilde{\theta} \sim \text{Uniform}[0, 1]$. Given $\tilde{\theta} = \theta$, the two flips $\tilde{x}_1$ and $\tilde{x}_2$ are conditionally independent Bernoulli random variables with parameter θ , where 1 denotes heads.

- (a) Compute the joint pmf of $\tilde{x}_1$ and $\tilde{x}_2$.
- (b) Both flips are observed to be heads. Compute $\mathbb{P}(\tilde{\theta} < 1/2 \mid \tilde{x}_1 = 1, \tilde{x}_2 = 1)$.
- (c) Are the two flips independent before conditioning on $\tilde{\theta}$? Use your answer to part (a) to justify your conclusion.

Problem 4 (backup): True or false?

Optional if time allows. Suggested working time: 5–7 minutes.

- (a) For a jointly Gaussian vector, two coordinates with covariance zero are independent.
- (b) If every observation has the same known covariance matrix, changing that matrix changes the ML estimate of the Gaussian mean vector.
- (c) In a Bayesian model, the posterior is proportional to the prior divided by the likelihood.
- (d) Random variables may be conditionally independent given a shared parameter but dependent after that parameter is integrated out.